A278299: Smallest Complete Polyomino Coloring

Minimum cells in a polyomino with n-coloring where every color pair shares an edge

Proved values: $a(12) = 40$, $a(13) = 46$, $a(14) = 56$, $a(15) = 61$, $a(16) = 69$, $a(17) = 77$, $a(18) = 90$, $a(19) = 97$, $a(20) = 108$, $a(21) = 119$

Computed by Peter Exley, March 2026

$a(12) = 40$ **[PROVED]** 5×8 bounding box, 66 color pairs. Edge bound: $65 < 66$

F	D	C	K	D	J	H	B
L	A	E	F	H	I	G	D
D	I	L	J	A	B	L	C
E	K	H	E	G	J	K	B
B	G	C	I	F	C	A	F

$a(13) = 46$ **[PROVED]** 7×7 bounding box, 78 color pairs. CP-SAT UNSAT: 11 shapes infeasible

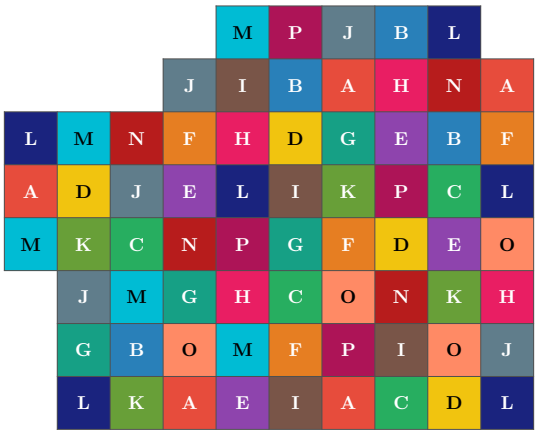
	A	J	L	M	J	
I	F	B	A	D	C	L
A	C	M	G	L	B	H
M	K	F	J	E	D	K
I	B	G	D	H	F	L
K	E	C	I	G	E	I
	M	H	J	K	A	H

$a(14) = 56$ **[PROVED]** 7×8 bounding box, 91 color pairs. Contact bound: $12 < 13$

G	F	L	B	A	D	F	B
M	E	G	N	M	L	C	M
J	B	I	J	K	A	H	F
E	K	D	M	I	F	K	N
I	C	B	H	L	J	G	C
K	A	G	D	E	A	H	J
L	N	E	C	H	I	N	D

$a(15) = 61$ **[PROVED]** 7×9 bounding box, 105 color pairs. Edge bound: $104 < 105$

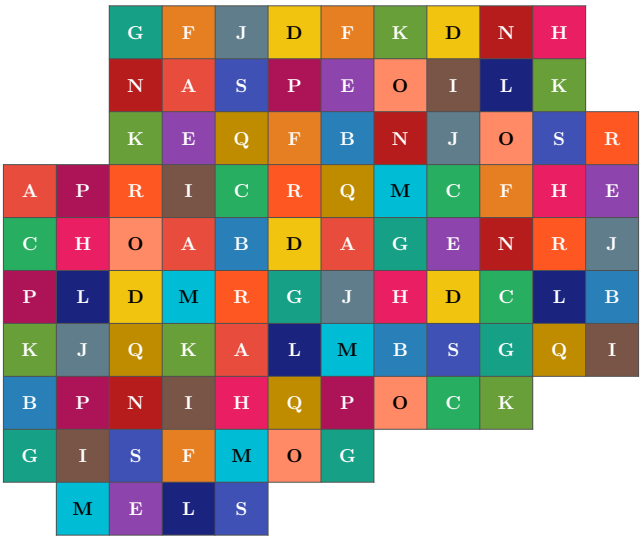
$a(16) = 69$ **[PROVED]** 8×10 bounding box, 120 color pairs. Edge bound: $119 < 120$



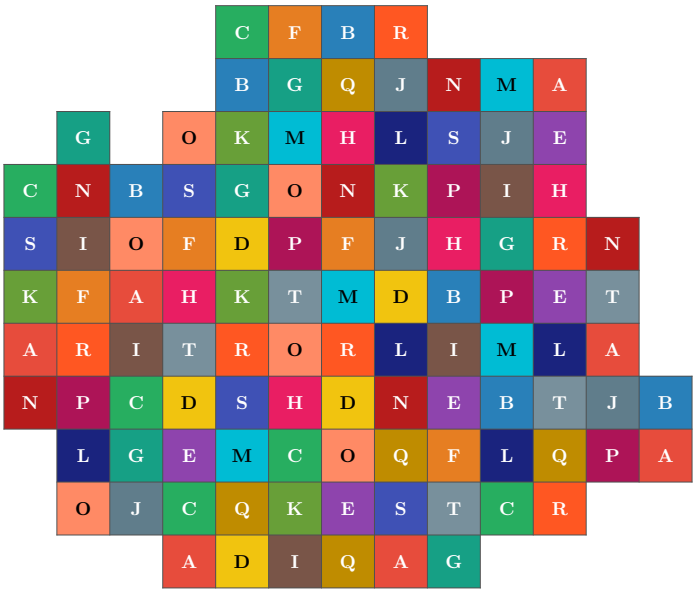
$a(17) = 77$ **[PROVED]** 8×10 bounding box, 136 color pairs. CP-SAT UNSAT: 28 shapes infeasible

$a(18) = 90$ **[PROVED]** 9×10 bounding box, 153 color pairs. Contact bound: $16 < 17$

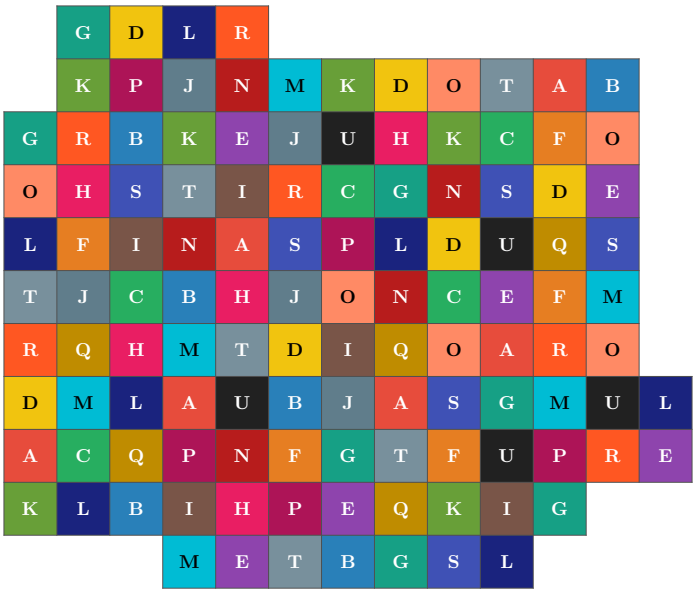
$a(19) = 97$ **[PROVED]** 10×12 bounding box, 171 color pairs. CP-SAT zero-slack: 28 shapes infeasible



$a(20) = 108$ **[PROVED]** 11×13 bounding box, 190 color pairs. Interior edge counting: $85 > 78$



$a(21) = 119$ **[PROVED]** 11×13 bounding box, 210 color pairs. Interior edge sum: $144 > 142$



Color Legend

A=0 B=1 C=2 D=3 E=4 F=5 G=6
H=7 I=8 J=9 K=10 L=11 M=12 N=13
O=14 P=15 Q=16 R=17 S=18 T=19 U=20